Mach's mechanics – the critique of absolute space

In 1883 the Austrian physicist Ernst Mach published *Die Mechanik in ihrer Entwicklung historisch-kritisch dargestellt* – "The Science of Mechanics: A Critical and Historical Account of its Development". The book traced mechanics from Archimedes to Hamilton and asked each time: what has actually been *observed*? Mach's answer to Newton's bucket experiment became one of the ideas that led the young Einstein to seek a new theory of gravity. Here are the book's main ideas, Mach's principle and the road to the measurements of Gravity Probe B – with four interactive 3D simulations.

Ernst Mach (1838–1916)

Ernst Mach was born on 18 February 1838 in Chirlitz (Chrlice) near Brünn in Moravia and died on 19 February 1916 in Vaterstetten near Munich. He was professor of experimental physics in Prague for almost three decades and from 1895 professor of the history and theory of the inductive sciences – in practice philosophy – in Vienna. He was an unusually broad scientist:

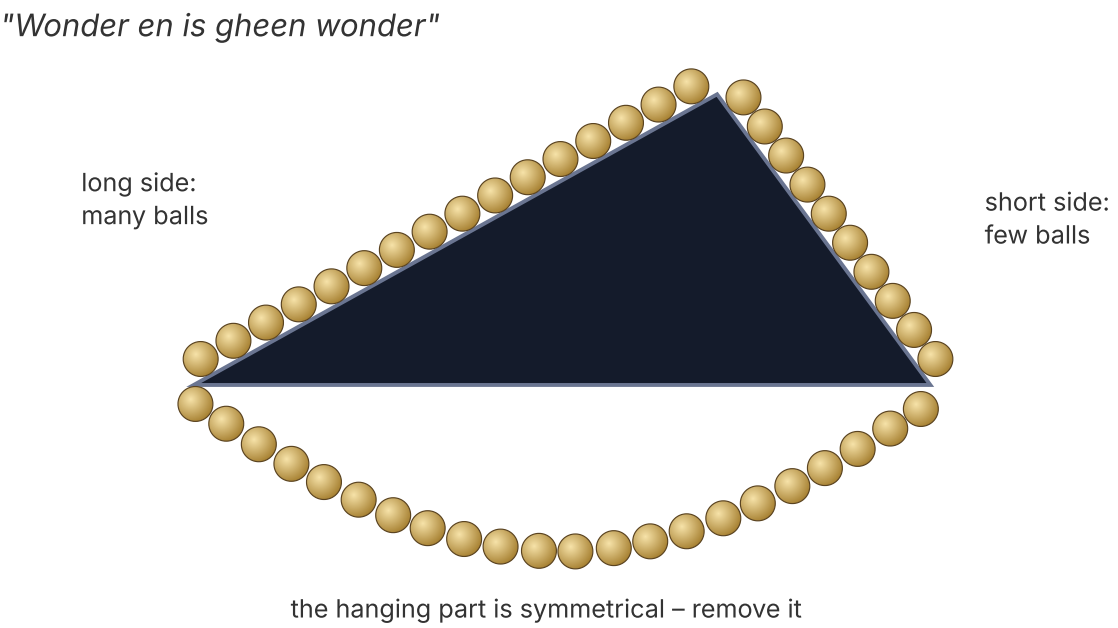
- **Physiology of perception:** in 1865 he described the light and dark stripes the eye sees at a boundary between two shades of grey, although they are not in the image – today called *Mach bands*.
- **Shock waves:** in 1886–87, together with Peter Salcher, he photographed a projectile flying faster than sound and showed the conical shock wave for the first time. The ratio of a speed to the speed of sound is therefore today called the *Mach number* (the name was introduced by Jakob Ackeret in 1929).
- **Theory of knowledge:** according to Mach, science should describe sensations as *economically* as possible (*Denkökonomie*). Concepts that cannot be traced back to something observable are metaphysical baggage – including Newton's absolute space and time and, he thought, the atoms.

The book – and Stevin's chain

Die Mechanik appeared in 1883 and in seven editions during Mach's lifetime; the English translation, *The Science of Mechanics*, came in 1893. Mach did not write a textbook but a critical history: how did Archimedes, Stevin, Galileo, Huygens and Newton arrive at the fundamental laws, and what did they really rest on? The book was read by a whole generation of physicists. Einstein read it as a student in Zurich around 1897 on the advice of his friend Michele Besso, and wrote in his obituary of Mach in 1916 that it had made a deep impression on him.

Mach's favourite example was the Flemish engineer Simon Stevin's proof of 1586 for the law of forces on an inclined plane. Lay a closed chain of identical balls over a triangle with two slopes of different length. If the chain started to move, it would go on for ever, because after a while it looks exactly as before – a perpetual-motion machine. So it stands still. The hanging part below the triangle is

symmetrical and can be removed, and then the balls on the two slopes must hold each other in equilibrium. The number of balls is proportional to the length of each slope, so the force along an inclined plane is proportional to $\sin\alpha$:



Stevin's chain (*clootcrans*) from *De Beghinselen der Weeghconst*, 1586. Stevin put it on the title page with the words "A wonder, and yet no wonder".

For Mach the proof was a model: it rests not on a law but on the experience that perpetual-motion machines do not exist. All of mechanics ultimately rests on such experiences, and nothing can be derived from pure thought.

Mach's definition of mass

Newton defined mass in the Principia as "quantity of matter", density times volume. Mach pointed out in 1868 that this is circular: density can only be defined as mass per volume. Instead he proposed a definition that uses only something measurable – accelerations. Let two bodies act on each other and only on each other (collisions, a spring, gravity). Experience shows that their accelerations are always opposite, and that the ratio between them is the same however they act on each other. One then *defines* the ratio of the masses as

$$\frac{m_A}{m_B} = -\frac{a_B}{a_A} = -\frac{\Delta v_B}{\Delta v_A}.$$

The definition is useful only because experience shows one more thing: it is **transitive**. If one has measured m_A/m_B and m_B/m_C, the product gives exactly the ratio m_A/m_C measured directly. Only for that reason can each body be given a single number, its mass, relative to a chosen standard. Newton's third law then becomes a definition, and the conservation of momentum a fact of experience.

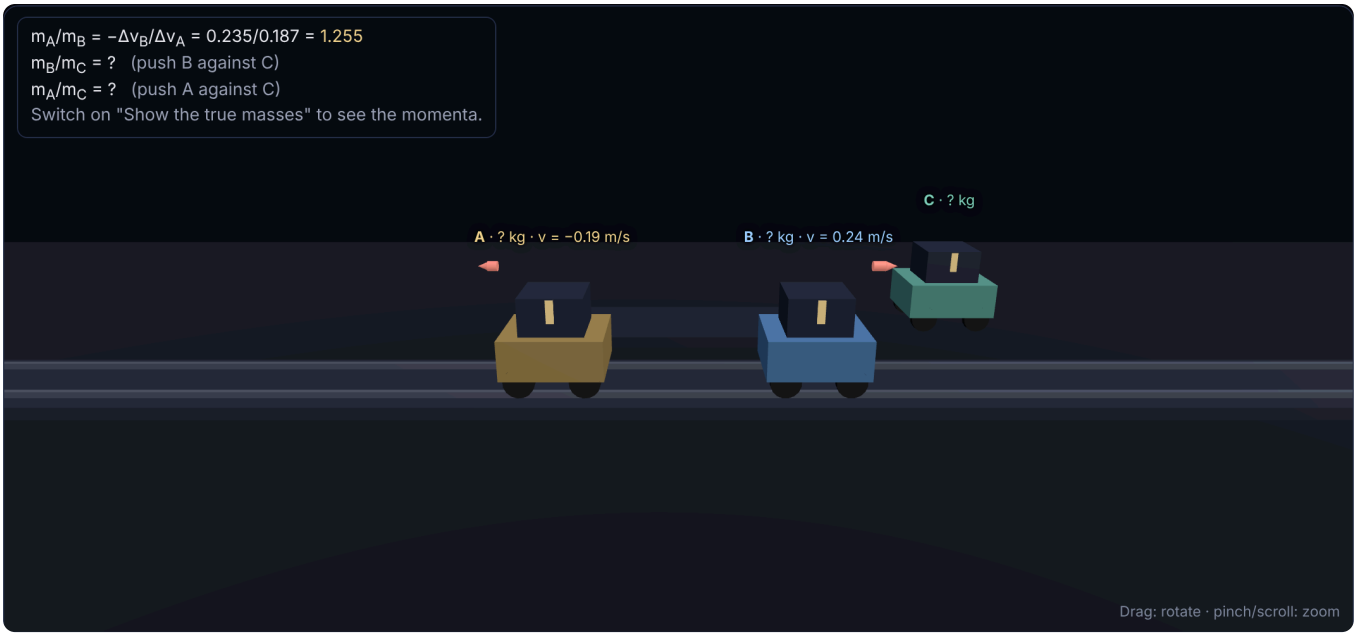


Figure 1. Mach's definition of mass – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/mach-mekanik-en.html#masse

Try it yourself. Three trolleys have unknown masses. A compressed spring between two of them is released, and the trolleys fly apart. The table at the top shows the ratio of the speeds.

1. Measure m_A/m_B and m_B/m_C. Predict m_A/m_C, and then measure it directly.
2. Is $p_A + p_B = 0$ after each push? What has that to do with Newton's third law – and what here is definition and what is experience?
3. Why do we not need a kilogram weight to measure the ratios?

Newton's bucket – and Mach's reply

In his scholium on space and time in the Principia, Newton described an experiment. Hang a bucket of water from a rope, twist the rope and release the bucket. At first the bucket turns while the water stands still, and the surface is flat. Then friction slowly sets the water rotating, and the surface becomes concave. If the bucket is stopped, the water goes on turning and is still concave. So the curvature is not linked to the water's motion relative to the bucket – it is greatest when the relative motion is least. For Newton this was proof of an absolute rotation relative to an absolute space.

Mach replied in *Die Mechanik*:

> *Newton's experiment with the rotating vessel of water simply informs us that the relative rotation of the water with respect to the sides of the vessel produces no noticeable centrifugal forces, but that such forces are produced by its relative rotation with respect to the mass of the earth and the other celestial bodies. No one is competent to say how the experiment would turn out if the sides of the vessel increased in thickness and mass till they were ultimately several leagues thick.*

Mach did not claim to know the answer. He claimed that Newton could not know it, because the experiment had never been made with masses other than the bucket's. All one can observe is rotation *relative to something*: to the bucket, to the Earth, to the fixed stars. And the fixed stars are enough to explain the experiment. Try switching between the two points of view in the simulation. Seen from the

water, the water stands still and the whole starry sky turns round – and yet the surface is concave. Is it the water's rotation in space or the sky's rotation about the water that curves the surface? According to Mach, these are two descriptions of the same thing.

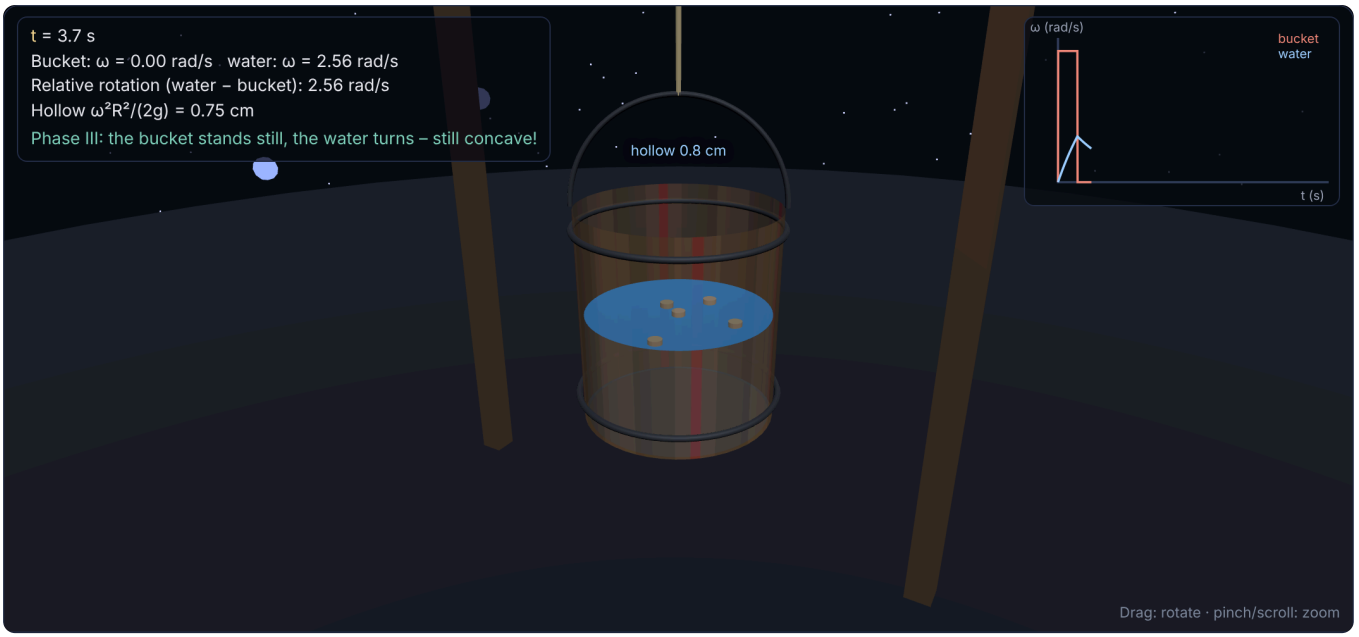


Figure 2. Newton's bucket – and Mach's reply – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/mach-mekanik-en.html#spanden

Try it yourself. The graph shows the angular velocity of the bucket and of the water. The red marks are on the bucket, the light corks float on the water.

1. Release the bucket. When is the relative rotation between water and bucket greatest – and when is the surface most concave?
2. Stop the bucket. Now the relative rotation is large again. Why does the surface not become flat?
3. Switch to the water's frame. What does the water "see"? Can one decide, from what the water sees, whether it is the water or the universe that turns?
4. The depth of the surface is $\omega^2 R^2/(2g)$. Does it match the number in the readout?

Foucault's pendulum – why does it follow the stars?

In 1851 Léon Foucault hung a 28 kg bob on a 67 m wire under the dome of the Panthéon in Paris. The plane of the pendulum's swing turned slowly clockwise – the Earth was turning beneath it. At the North Pole the plane turns once round in a sidereal day (23 h 56 min); at latitude φ it takes

$$T = \frac{23.93\ \mathrm{h}}{\sin\varphi},$$

about 32 hours in Paris, about 29 hours in Grenaa in Denmark, and at the equator the plane does not turn at all. Mach asked: *relative to what* does the pendulum keep its plane fixed? The answer is, to high accuracy, the distant stars and galaxies. A pendulum, a gyroscope and an inertial navigation system without a compass (see the gyroscope) all point to the same "inertial" frame, and that frame

stands still relative to the masses of the universe. Is that a coincidence – or does the mass of the universe determine what is inertial? That is the heart of Mach's principle.

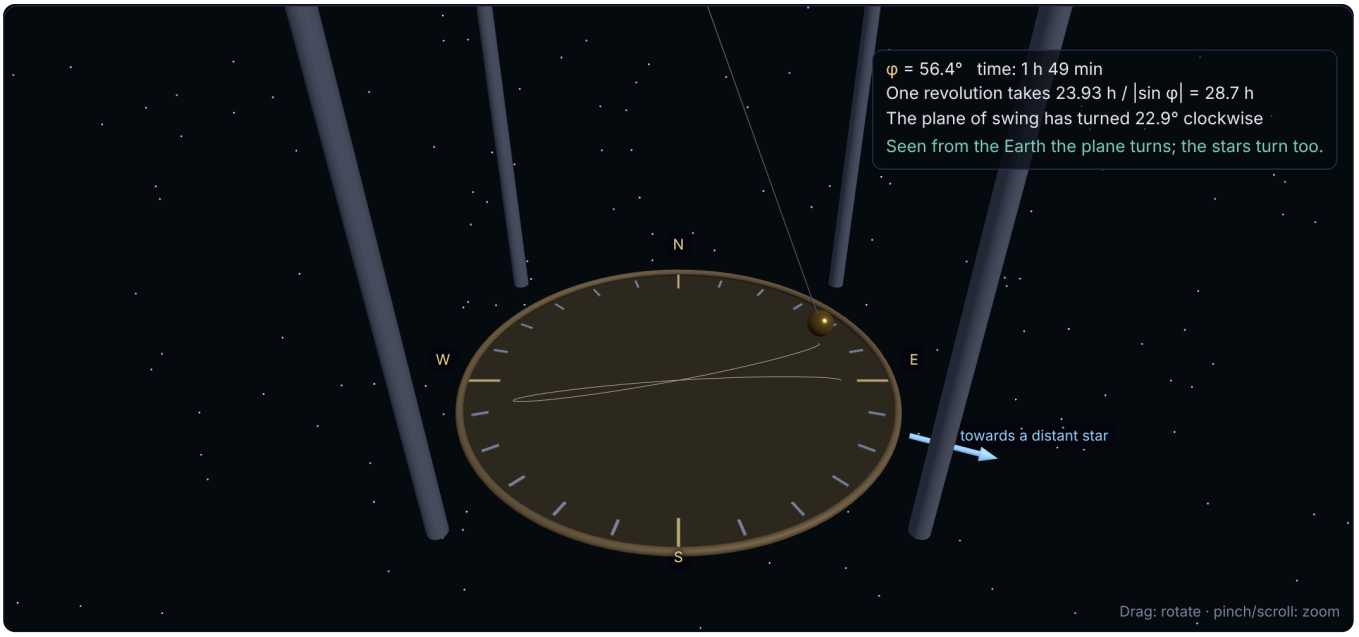


Figure 3. Foucault's pendulum – why does it follow the stars? – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/mach-mekanik-en.html#foucault

Try it yourself. The pendulum draws its trace in the sand on the floor. Time is speeded up so that a day passes in a few minutes.

1. Choose Grenaa. How many hours pass before the plane of swing has turned half a revolution?
2. Switch to the stars' frame. What is turning now – the pendulum or the floor?
3. Choose the equator and Sydney. What happens, and why does the plane turn the opposite way in the southern hemisphere?

Mach's principle – and Einstein

Mach never formulated a precise principle. He merely said that inertia and the fictitious forces in a rotating frame *could* be due to the other masses of the universe, and that a physics speaking only of relative motions would be preferable. The young Einstein took the idea seriously. On 25 June 1913, in the middle of his work on general relativity, he wrote from Zurich to Mach that in his new theory inertia would arise from an interaction between bodies, entirely in the spirit of Mach's bucket considerations. If a heavy shell of matter were rotated relative to the fixed stars, a weak Coriolis force would arise inside the shell, and a Foucault pendulum would be dragged round.

In 1918, in the paper "Prinzipielles zur allgemeinen Relativitätstheorie", Einstein gave the idea the name **Mach's principle**: the metric of space – and thereby inertia – should be completely determined by the mass in the universe. It turned out that his own theory does not quite live up to it:

- Willem de Sitter found in 1917 a solution of the field equations *without* matter, in which test particles nevertheless have inertia.
- Kurt Gödel found in 1949 a universe in which all the matter rotates relative to the local inertial frames – something Mach's principle should forbid.

- On the other hand, general relativity contains exactly the effect Einstein described in his letter: **frame dragging**. A rotating mass drags the local inertial frames with it.

Einstein gradually lost faith in the principle. But the question of why the inertial frames stand still relative to the distant galaxies is still not settled, and it is discussed today by, among others, Julian Barbour.

Thirring's rotating shell – and Gravity Probe B

In 1918 the Austrian physicist Hans Thirring calculated Einstein's thought experiment. A thin spherical shell of mass M and radius R, rotating with angular velocity ω, makes the inertial frames inside the shell rotate with

$$\Omega = \frac{4}{3}\,\frac{GM}{c^2 R}\,\omega .$$

The formula holds when the shell is light, $GM/(c^2R) \ll 1$. In the same year Josef Lense and Thirring calculated the corresponding effect for a rotating planet: the **Lense–Thirring effect**. For the Earth $GM/(c^2R) \approx 7 \times 10^{-10}$, so the effect is tiny. Even so, it has been measured. The satellite *Gravity Probe B* flew in 2004–2005 with four superconducting gyroscopes in a polar orbit at 642 km altitude. The result was published in 2011:

Effect	Measured (Gravity Probe B, 2011)	General relativity
Geodetic precession (curvature of spacetime)	6601.8 ± 18.3 milliarcseconds per year	6606.1
Frame dragging (Lense–Thirring)	37.2 ± 7.2 milliarcseconds per year	39.2

37 milliarcseconds per year is the angular size of a human hair seen from 400 metres away. What if the shell were the whole universe? Dieter Brill and Jeffrey Cohen showed in 1966 that when the shell's radius approaches its own Schwarzschild radius, the dragging becomes complete, $\Omega \to \omega$: inside, one can no longer distinguish the shell's rotation from one's own. For the visible universe $GM/(c^2R)$ is indeed of order 1 – a striking coincidence that Dennis Sciama tried in 1953 to turn into a theory of inertia.

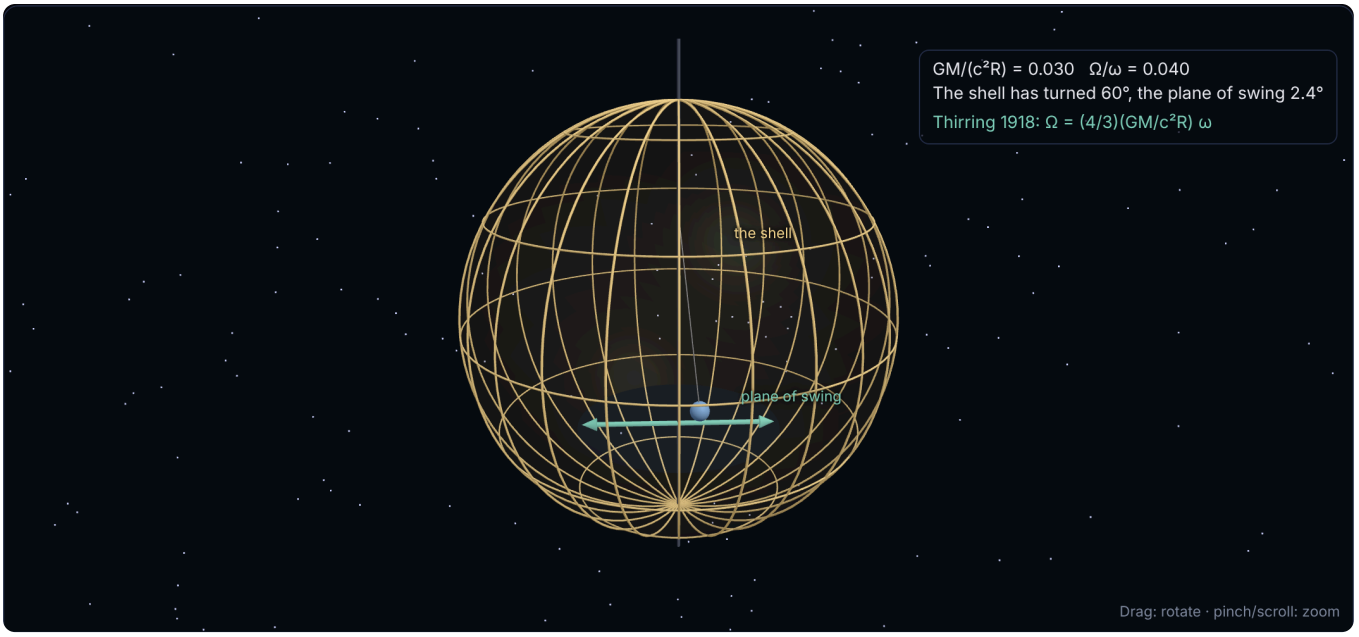


Figure 4. Thirring's rotating shell – and Gravity Probe B – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/mach-mekanik-en.html#thirring

Try it yourself. The golden shell turns about the vertical axis. In the middle a Foucault pendulum swings; its plane of swing is the local inertial frame.

1. Set the compactness to 0. What does the pendulum's plane do? Turn it up – how fast does the plane follow the shell?
2. Choose "The Earth". How many years must one wait before the plane has turned one degree?
3. Choose "The whole universe". Can the pendulum now decide whether it is the shell or the pendulum that turns? What would Mach say?

Mach against the atoms – and against Einstein?

Mach's demand that physics speak only of the observable made him one of the last great opponents of atomic theory. According to an often-repeated anecdote, after a lecture by Ludwig Boltzmann in Vienna in 1897 he asked: "Haben Sie eins gesehen?" – "Have you seen one?" Boltzmann, who built all of statistical mechanics on atoms (see entropy and the arrow of time), felt isolated and died in 1906, shortly before atoms became generally accepted. Max Planck attacked Mach sharply in 1908–1910. The decision came by a route Mach had not foreseen: Einstein showed in 1905 how the impacts of atoms can be seen as the Brownian motion of small particles in water, and Jean Perrin's measurements in 1908 gave Avogadro's number. The atoms were now observable.

Special relativity, which rejects absolute time and absolute simultaneity precisely as unobservable, is very much in Mach's spirit, and Einstein himself called Mach a forerunner. Yet the preface to Mach's *Die Prinzipien der physikalischen Optik*, published posthumously in 1921, contains a sharp rejection of relativity. The philosopher Gereon Wolters has since argued that the preface was written by Mach's son Ludwig – who himself tried to disprove relativity – and not by Mach. The question is still debated.

Legacy

- **The Vienna Circle:** the society of logical positivism in Vienna was called from 1928 *Verein Ernst Mach*.
- **Einstein:** in his obituary of 1916 Einstein wrote that Mach's critique of Newton had been decisive for him. Mach's principle was one of the three principles on which his general theory of relativity was built.
- **Quantum mechanics:** Heisenberg's demand in 1925 to build the theory solely on observable quantities was Mach's programme – even though it was about the atoms that Mach had rejected.
- **Modern physics:** frame dragging has been measured with Gravity Probe B and the LAGEOS satellites and is seen in matter around rotating black holes. Whether inertia itself is due to the mass of the universe is still an open question.

Exercises

1. Two trolleys are pushed apart by a spring. Trolley A gets the speed 0.30 m/s, trolley B 0.45 m/s in the opposite direction. Find m_A/m_B. If $m_B = 2.0$ kg, what is m_A?
2. Show that Mach's definition is transitive if momentum is conserved in all collisions. Why is the transitivity a fact of experience and not a logical necessity?
3. A bucket of radius 15 cm rotates at 2 revolutions per second. How deep is the hollow of the water surface? (The surface is the paraboloid $z = \omega^2 r^2/(2g)$.)
4. How long does it take a Foucault pendulum in Grenaa ($\varphi = 56.4^\circ$) to turn through 90°?
5. Use Thirring's formula: how massive must a shell of radius 1 m be before the pendulum inside it turns 1 % as fast as the shell? Compare with the mass of the Earth.
6. Gravity Probe B measured a frame dragging of 37 milliarcseconds per year. How many years would it take before the gyroscope had turned 1°?

Sources and further reading

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